

MT 1008 MVC

Book: Thomas Calculus 14

Quiz: 8% Assignment: 7% Homework: 5%

Sessionals: (I + II) 30% Final: 50%

## Chapter #14

## Partial Derivatives

### { 14.1 Function of Several Variables

$$f: A \rightarrow B$$

$$f: \mathbb{R} \rightarrow \mathbb{R}$$

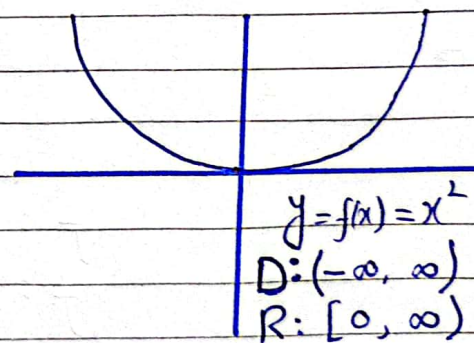
$y = f(x)$  real valued function of single variable

Input variable/Independent  
Dependent Variable/output

collection of

$$(x, y) \rightarrow \mathbb{R} \times \mathbb{R}$$

Graph of  $\mathbb{R}^2$  is entire  
x-y plane



$x \in \mathbb{R}$   
i)  $x > 0$  ii)  $x < 0$  iii)  $x = 0$

$$\mathbb{R}^2: \mathbb{R} \times \mathbb{R}$$

$U \in \mathbb{R}^2$  Euclidean Space

$$U: (a, b)$$

$U \in \mathbb{R}^3 \rightarrow$  constitute the entire 3D space

First octant

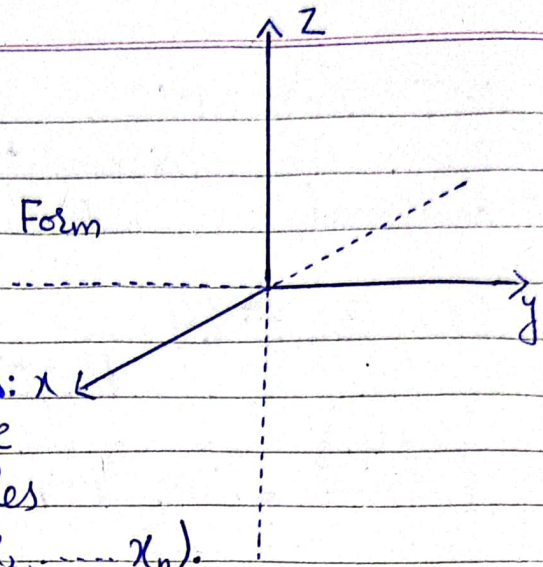
$x: +ve, y: +ve, z: +ve$



$$w \in \mathbb{R}^4$$

$$w : (a, b, c, d)$$

$$w \in \mathbb{R}^n \text{ General Form}$$



**Function of Several Variables:**

Suppose

$D$  is set of  $n$ -tuples of real numbers  $(x_1, x_2, \dots, x_n)$ .

A real valued function  $f$  on  $D$  is a rule that assigns a unique (single) number

$$w = f(x_1, x_2, \dots, x_n)$$

to each element of  $D$ . The set  $D$  is function's domain. The  $w$ -values taken on by  $f$  is function's range. The  $w$  is dependent variable and  $f$  is said to be function of  $n$ -independent variable  $(x_1, x_2, \dots, x_n)$ .

→ We also call  $x_j$  ( $j=1, \dots, n$ ) the input variable and  $w$  is output variables.

$$f : D \rightarrow \mathbb{R}$$

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}$$

$w = f(x_1, x_2)$  real valued function of two variables.

$$z = f(x, y).$$

$$(x, y, z) = (x, y, f(x, y))$$

↳ Graphed in  $\mathbb{R}^3$   
Collection of all possible tuples.



$$f: \mathbb{R}^3 \rightarrow \mathbb{R}$$

$$w = f(x_1, x_2, x_3)$$

$$w = f(x, y, z)$$

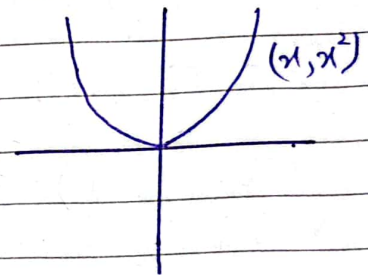
$$(x, y, z, w) \in \mathbb{R}^4$$

- Function of one variable can be drawn on  $\mathbb{R} \times \mathbb{R}$ .
- Function of two variables can be drawn on  $\mathbb{R}^2 \times \mathbb{R}$  while not for 3 variables.

Domain will be some part of  $xy$ -plane (can be entire  $xy$ -plane) for two variable function.

- For every ordered pair there is a unique element of  $\mathbb{R}$

$$y = f(x) = x^2$$



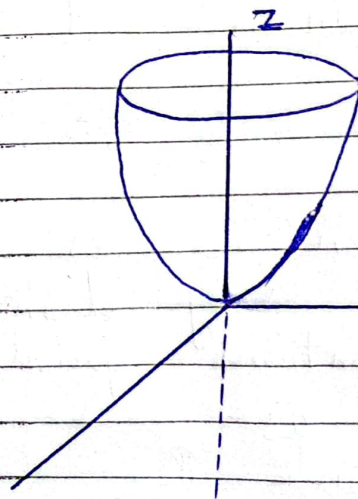
$$z = f(x, y) = x^2 + y^2 \text{ (Paraboloid)}$$

Domain = Entire  $xy$ -plane  
Range:  $[0, \infty)$

$$w = f(x, y, z) = x^2 + y^2 + z^2$$

Domain:  $\mathbb{R}^3$

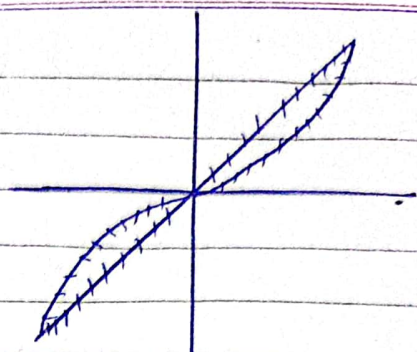
Range:  $[0, \infty)$



$$z = f(x, y) = \frac{(x-1)(y+2)}{(y-x)(y-x^3)}$$



$$\begin{array}{ll} (1, 1) & (2, 8) \\ (2, 2) & (3, 27) \\ (3, 3) & \vdots \\ \vdots & \vdots \end{array}$$

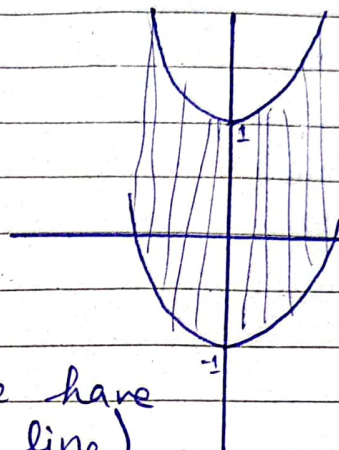


Ex # 14.1

Q9)  $f(x, y) = \cos^{-1}(y - x^2)$

$$-1 \leq y - x^2 \leq 1$$

$$-1 + x^2 \leq y \leq 1 + x^2$$



Function of single variable have intervals (sub set of real line) as domain while for two variables have part of regions sketched.

**Interior Points:**

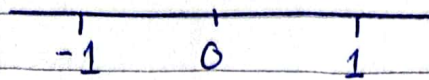
A point  $(x_0, y_0)$  in the (set) region  $\mathbb{R}$  in  $xy$ -plane is an interior point of  $\mathbb{R}$  if it is center of a disk with +ve radius, that lies entirely in  $\mathbb{R}$ .

**Boundary Point:**

A point  $(x_0, y_0)$  of region  $\mathbb{R}$  is boundary point of  $\mathbb{R}$  if every disk centered at  $(x_0, y_0)$  contains points that lie outside  $\mathbb{R}$  as well as points that lie inside  $\mathbb{R}$ .



$$|x| < 1$$



$$|x - a| < R$$

**Open Region:**

A region is open if it consists entirely of its interior points.

**Closed Region:**

A region is closed if it contains all its boundary points.

**Bounded Region:**

A region in plane ( $\mathbb{R}^2$ ) is bounded if it lies within the disk of finite radius.

- A region in space is bounded if it lies within the disk of finite radius.

|                                           |   |                                           |   |
|-------------------------------------------|---|-------------------------------------------|---|
| $z = f(x, y)$<br>domain in $\mathbb{R}^2$ | ✓ | $w = f(x, y, z)$<br>domain $\mathbb{R}^3$ | ✗ |
|-------------------------------------------|---|-------------------------------------------|---|

**Q #10**

$$f(x, y) = \ln(xy + x - y - 1)$$

$$xy + x - y - 1 > 0$$

$$xy + x - y - 1 > 0$$

$$x(y+1) - 1(y+1) > 0$$

$$(x-1)(y+1) > 0$$

Case 1

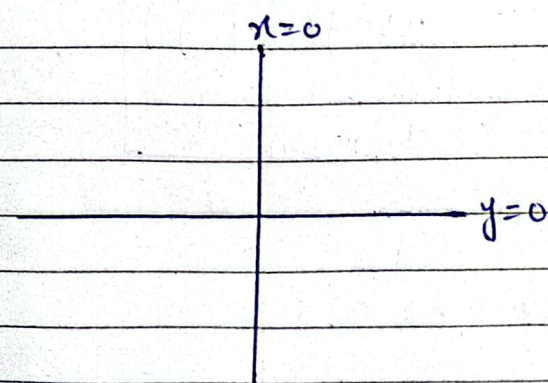
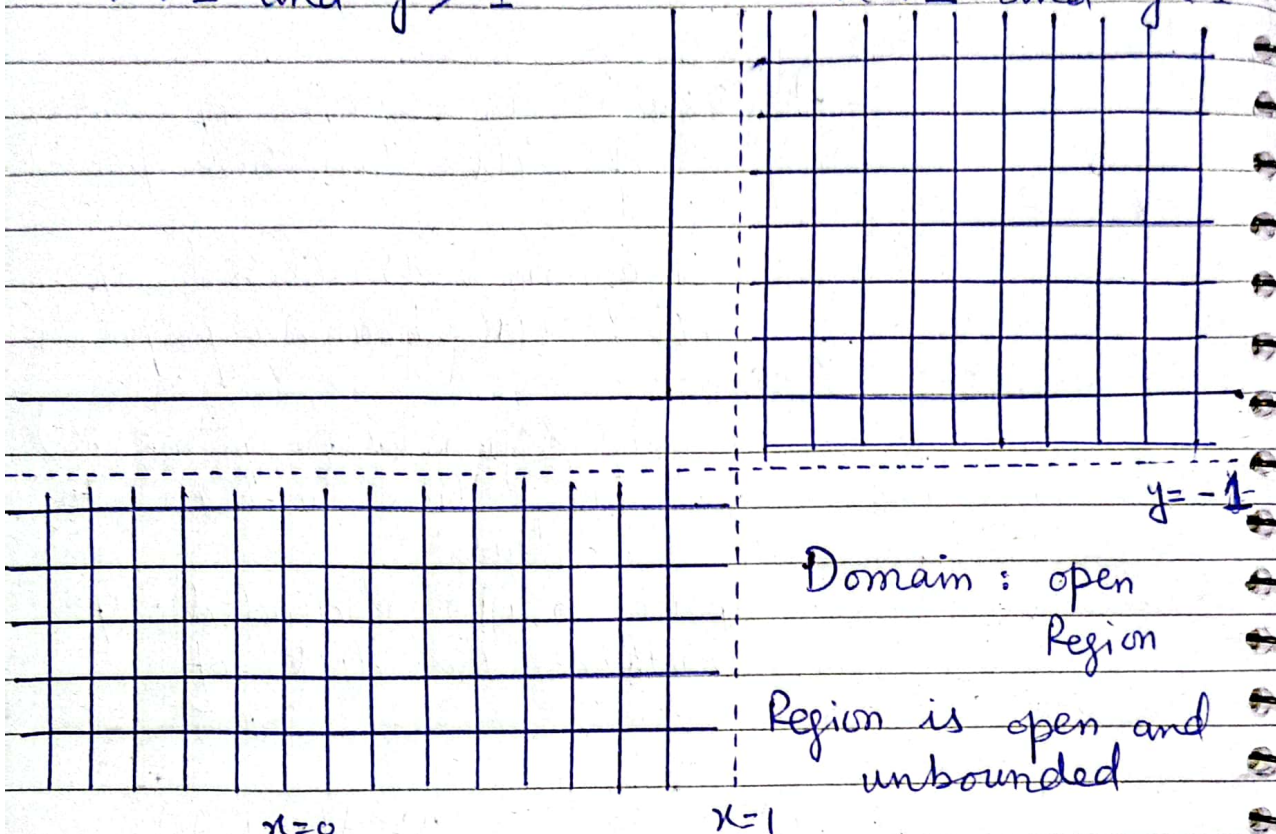
$$\begin{aligned} x-1 &> 0 \\ \text{and} \\ y+1 &> 0 \end{aligned}$$

$$x > 1 \text{ and } y > -1$$

Case 2

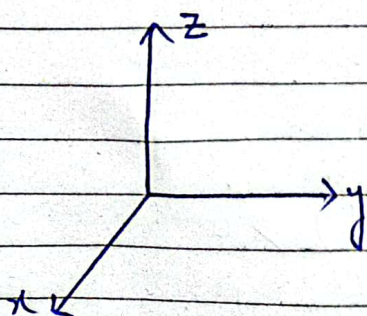
$$\begin{aligned} x-1 &< 0 \\ \text{and} \\ y+1 &< 0 \end{aligned}$$

$$x < 1 \text{ and } y < -1$$



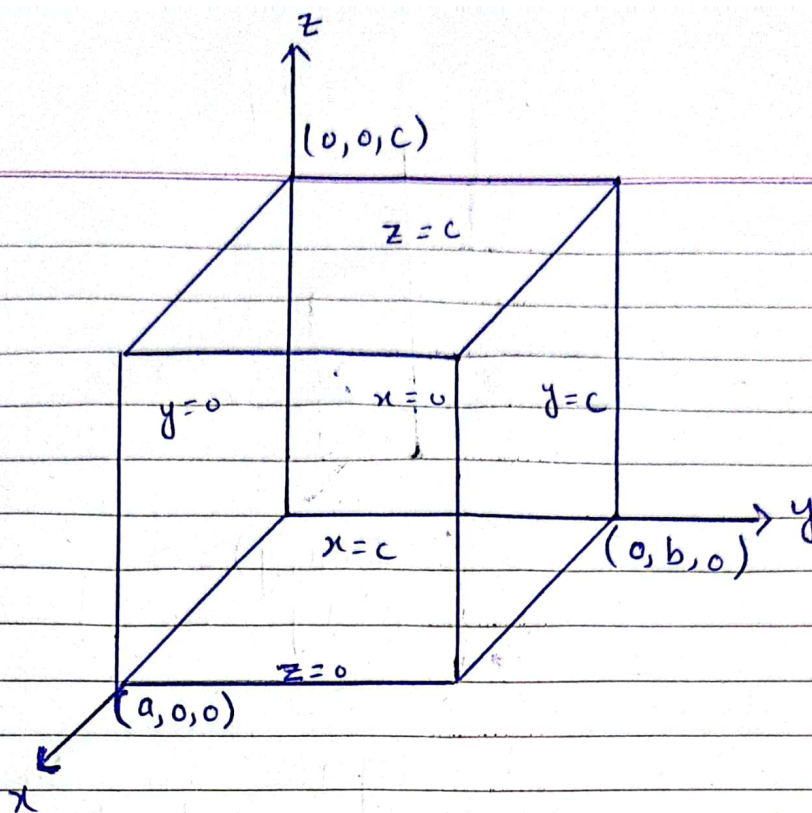
$\left. \begin{aligned} x=0 \\ y=0 \end{aligned} \right\}$  co-ordinate lines

$\left. \begin{aligned} x=a \\ y=b \end{aligned} \right\}$  lines || to co-ordinate lines



$\left. \begin{aligned} z=0, & \text{ xy-plane} \\ x=0, & \text{ yz-plane} \\ y=0, & \text{ xz-plane} \end{aligned} \right\}$  co-ordinate planes





### Level Curve:

A curve in plane where function  $f(x, y)$  has constant value  $f(x, y) = C$  is called level curve. The set of points  $(x, y, f(x, y))$  in space, for  $(x, y)$  in the domain of  $f(x, y)$ , is called graph of  $f(x, y)$ . This is also called surface  $z = f(x, y)$ .

### Contour Curve:

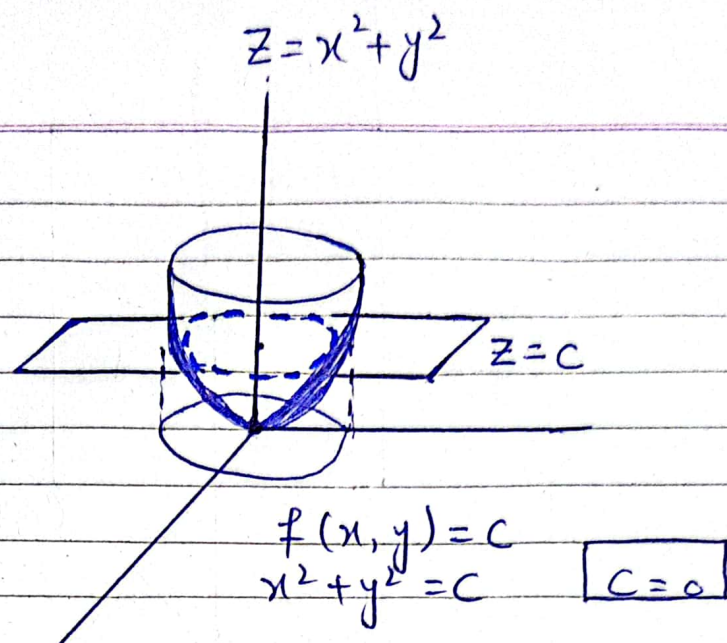
A curve in space in which plane  $z = C$  intersect the surface  $z = f(x, y)$  is made up of points that represent the function value  $f(x, y) = C$ .

It is called contour curve  $f(x, y) = C$  to distinguish it from the level curve  $f(x, y) = C$  in the domain of  $f(x, y)$ .

$$z = f(x, y)$$

$$z = C$$

$$f(x, y) = C$$



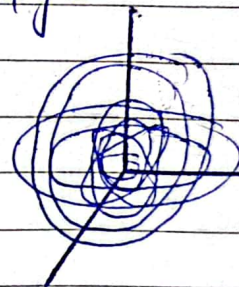
### Level Surface:

A surface in the space where  $f(x, y, z)$  has constant value  $f(x, y, z) = C$  is called level surface.

$$w = f(x, y, z) = x^2 + y^2 + z^2$$

$$w = C$$

$$x^2 + y^2 + z^2 = C$$



Q # 51:

$$f(x, y) = \sqrt{x + y^2 - 3} \quad (3, -1)$$

$$f(x, y) = C$$

$$\sqrt{x + y^2 - 3} = C$$

$$\sqrt{3 + 1 - 3} = C$$

$C = 1$

$$x + y^2 - 3 = 1$$

$$y^2 = 4 - x$$

